ProGrade: A Medallion Architecture-Driven Early Warning Framework for Academic Risk Prediction Using Ensemble Learning and Temporal Checkpoints

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***Abstract*—Early identification of at-risk students remains a persistent challenge in higher education, where interventions are frequently applied after the window for effective remediation has closed. This paper presents ProGrade, an end-to-end early warning framework integrating a Medallion data warehouse architecture with temporal checkpoint modeling and explainable machine learning. Three public educational datasets are unified within a Kimball star-schema pipeline (Bronze, Silver, and Gold layers). A temporal feature-masking protocol progressively reveals features at simulated Weeks 4, 8, and 12, enabling systematic evaluation of the accuracy–lead time tradeoff. To ensure methodological validity, Mutual Information feature selection, SMOTE oversampling (k-neighbors$=5$), and feature scaling are applied exclusively within training folds, preventing data leakage. We benchmark Logistic Regression, Random Forest, and XGBoost across all checkpoints. At Week 12, XGBoost attains an F1-Macro of 0.852 ± 0.005 and ROC-AUC of 0.920 on held-out test data (five-seed stability analysis), while Week 8 models achieve F1-Macro exceeding 0.80, offering substantial lead time for counselor intervention. SHAP-based feature attribution provides global and individualized explanations, enabling transparent three-tier risk stratification supporting actionable, per-student intervention recommendations.**

***Index Terms*—Educational Data Mining, Student Dropout Prediction, Data Warehouse, Medallion Architecture, SHAP Explainability, Ensemble Learning, Early Warning Systems, Data Leakage Prevention.**

I. Introduction

Student attrition in higher education carries significant economic and social consequences for institutions, labor markets, and individuals alike [1]. Globally, dropout rates continue to strain institutional resources and deprive students of educational outcomes that are critical for long-term social mobility [2]. Despite these costs, the predominant institutional response remains *reactive*: students are identified as at-risk only after final grades are recorded, well beyond the point where targeted academic support could alter their trajectories.

Machine learning (ML) has emerged as a promising tool within Educational Data Mining (EDM) for predicting student outcomes [3], [4]. Ensemble methods—including Random Forest (RF) [5] and gradient-boosted trees such as XGBoost [6]—have consistently demonstrated strong predictive performance on structured educational data [7], [8]. However, the transition from research prototypes to deployable institutional tools is hindered by four recurring limitations in the existing literature:

1) **Absence of reproducible data architectures.** The majority of EDM studies apply ML directly to flat files or isolated datasets without formalized data governance [3]. Structured warehouse architectures following established dimensional modeling principles [9] are rarely adopted, despite being the industry standard for analytical workloads [10].
2) **Data leakage in preprocessing pipelines.** Many studies apply SMOTE, normalization, or imputation *prior* to train-test partitioning, leading to overstated performance metrics that do not generalize to unseen data [11], [12].
3) **Temporal mismatch between prediction and intervention.** Systems typically predict risk either at enrollment (using limited pre-admission features) or after semester completion (using outcome variables). The critical mid-semester window—when behavioral signals accumulate but final outcomes are not yet determined—remains underexplored [13], [14].
4) **Insufficient model interpretability.** Academic advisors require transparent, auditable explanations to justify targeted interventions. Opaque "black-box" predictions fail to establish the institutional trust necessary for adoption [15], [16].

To address these gaps, we introduce **ProGrade**—a production-oriented analytics framework that integrates a Medallion data warehouse with leakage-free ensemble modeling, temporal checkpoint evaluation, and SHAP-based explainability.

A. Research Questions

- **RQ1:** How does a structured Medallion warehouse architecture (Bronze $\rightarrow$ Silver $\rightarrow$ Gold) combined with temporal feature masking affect the reproducibility and predictive quality of EDM models?

- **RQ2:** At which temporal checkpoint (Week 4, 8, or 12) does the tradeoff between prediction accuracy and intervention lead time yield the most operationally viable early warning?
- **RQ3:** To what extent can SHAP feature-attribution explanations support transparent, per-student risk stratification for academic counselors?

B. Contributions

C1. Medallion Data Warehouse for EDM: We implement a three-layer Medallion Architecture (Bronze, Silver Star-Schema, Gold Data Mart) for educational analytics, establishing a reproducible and auditable data foundation.

C2. Temporal Checkpoint Protocol: We design a feature-masking protocol that simulates progressive information availability at Weeks 4, 8, and 12, enabling principled quantification of the accuracy–intervention lead time tradeoff.

C3. Leakage-Free ML Pipeline: We enforce strict post-split preprocessing—Mutual Information feature selection, SMOTE (k-neighbors=5), and StandardScaler applied exclusively to training data—yielding an F1-Macro of 0.852 ± 0.005 across five random seeds on held-out test data.

C4. Explainable Risk Stratification: We integrate SHAP TreeExplainer to generate both global feature importance summaries and per-student waterfall explanations, mapped to a three-tier risk classification (Low / Medium / High).

II. Related Work

A. Educational Data Mining and Student Outcome Prediction

The application of data mining to educational contexts has been extensively surveyed [3], [4]. Hellas et al. [8] systematically reviewed predictive modeling for academic performance, identifying feature engineering and model selection as primary determinants of accuracy. More recent surveys [7], [17] confirm that ensemble methods—particularly Random Forest and gradient boosting—consistently outperform linear models and neural approaches on tabular educational data. Umer et al. [18] provide a comprehensive 2024 review of dropout prediction methods, noting persistent gaps in temporal modeling and cross-institutional generalizability.

B. Data Governance in Educational Analytics

Despite the maturity of dimensional modeling in industry [9], educational analytics pipelines remain largely ad-hoc. Abella-García et al. [10] reviewed data warehouse architectures for education and identified a lack of standardized schemas as a barrier to institutional adoption. ProGrade addresses this gap by implementing a Medallion Architecture that separates raw ingestion, dimensional modeling, and analytical output into distinct layers with explicit data lineage.

C. Early Warning Systems

Early Warning Systems (EWS) for student risk have evolved from rule-based heuristics to ML-driven prediction. Arnold and Pistilli [19] presented Purdue's Course Signals, a pioneering system using simple traffic-light indicators. Jayaprakash et al. [20] extended this with open-source analytics, while Marbouti et al. [13] demonstrated early prediction at Week 4 using standards-based grading. However, these systems typically evaluate a single time point rather than systematically comparing prediction quality across multiple checkpoints. Aulck et al. [14] modeled dropout longitudinally across semesters but did not examine intra-semester temporal dynamics. Martins et al. [21] investigated phased prediction on the UCI Dropout dataset, providing a direct point of comparison for our temporal checkpoint protocol.

D. Data Leakage and Methodological Validity

Class imbalance is endemic to dropout prediction, with at-risk students typically constituting a minority class [11]. SMOTE [12] is the most widely adopted oversampling technique, but its misapplication before data splitting introduces synthetic samples derived from test data, contaminating model evaluation [3]. ProGrade enforces a strict separation: SMOTE and feature scaling are computed exclusively on the training partition, with validation and test sets transformed using training-derived parameters only.

E. Explainable AI in Education

Khosravi et al. [15] outlined the case for explainable AI (XAI) in education, arguing that transparency is a prerequisite for institutional adoption. Alwadi et al. [16] recently demonstrated SHAP-based early detection with transparent student support. Lundberg and Lee's [22] SHAP framework, grounded in cooperative game theory, provides both global and local feature attribution. ProGrade uses SHAP TreeExplainer to generate individualized waterfall plots that identify, for each flagged student, the specific features driving their risk classification.

F. Complementary Modeling Approaches

Recent work has explored alternative architectures for educational prediction. Graph Neural Networks (GNNs) and attention-based temporal models have been applied to capture relational and sequential structures in student interaction data [23]. Federated learning frameworks have also been proposed for privacy-preserving cross-institutional analytics, enabling collaborative model training without sharing raw student records [24]. These approaches address complementary challenges—relational structure, sequential dynamics, and data privacy—whereas ProGrade targets structured tabular data with production-grade data governance, temporal checkpoint evaluation, and full leakage prevention, representing a compatible but distinct design philosophy.

III. Methodology

ProGrade comprises four integrated components: (1) a Medallion data warehouse pipeline, (2) a temporal feature-masking protocol, (3) a leakage-free model training and evaluation pipeline, and (4) a SHAP-based explainability module. Fig. 1 illustrates the Silver layer star-schema design.

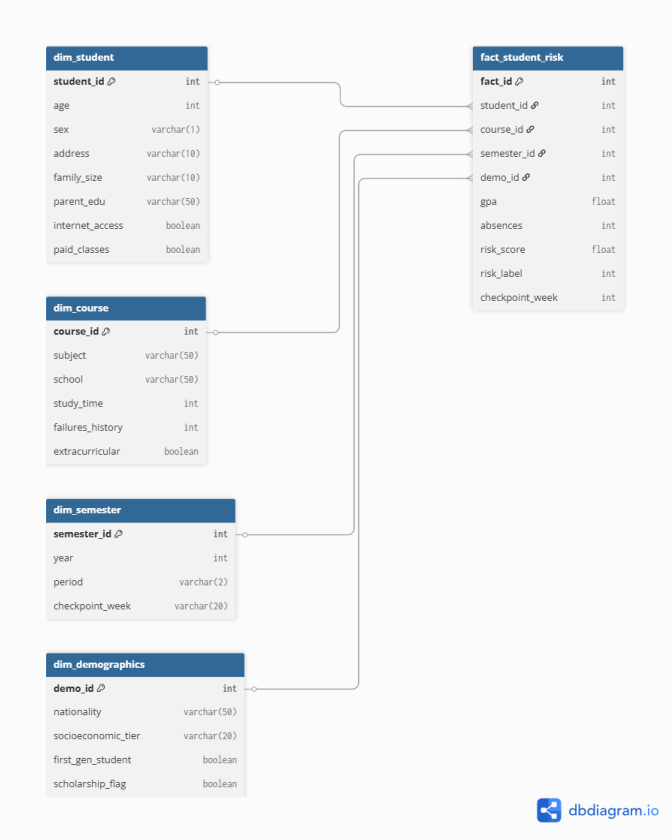


Fig. 1: ProGrade Silver layer star-schema. The central `FactStudentRisk` table links to four dimension tables (`DimStudent`, `DimCourse`, `DimSemester`, `DimDemographics`) via surrogate keys.

A. Datasets

Three publicly available educational datasets are unified within the ProGrade warehouse:

- **UCI Student Performance** [25]: 649 records of secondary school students in Portugal with demographic, social, and academic attributes. The binary risk label is derived as $G3 < 10$ (final grade below passing threshold).
- **UCI Dropout and Academic Success** [1], [26]: 4,424 records from a Portuguese polytechnic institution with 36 attributes spanning enrollment, demographic, academic, and macroeconomic dimensions. Risk label: `Target` = "Dropout". This is the primary modeling dataset due to its richness and temporal feature structure.
- **Kaggle Students Performance** [27]: 2,392 records with academic, behavioral, and parental attributes. Risk label: `GradeClass` ≥ 3 (grades D or F).

B. The Medallion Warehouse Pipeline

The ETL pipeline follows the Medallion Architecture pattern, decomposing data processing into three layers:

- **Bronze (Raw Ingestion):** Source datasets are ingested with path-integrity validation via declarative manifests. Raw files are stored without modification to preserve data provenance.
- **Silver (Dimensional Model):** Data is cleaned, imputed (median for numeric, mode for categorical), and mapped to a Kimball star-schema [9]. As shown in Fig. 1, the central `FactStudentRisk` table stores numeric measurements (GPA, absences, risk score) and links to four dimension tables via surrogate keys: `DimStudent` (personal/family attributes), `DimCourse` (subject/school context), `DimSemester` (temporal checkpoint), and `DimDemographics` (socioeconomic indicators). The warehouse is implemented in SQLite using SQLAlchemy ORM with indexed foreign keys.
- **Gold (Data Mart):** A denormalized analytical view (`student_risk_mart.csv`) joins all dimensions with human-readable mappings (e.g., "Urban"/"Rural", GPA brackets, attendance tiers) for direct consumption by BI dashboards and counselor reports.

C. Temporal Feature Masking

To simulate the progressive availability of information during a semester, we define three temporal checkpoints with distinct feature masks applied to the UCI Dropout dataset:

- **Week 4 (Enrollment):** Only demographic and pre-enrollment features are available: marital status, application attributes, parental qualifications and occupations, admission grades, age, gender, scholarship status, and nationality (21 features).
- **Week 8 (Mid-Semester):** First-semester academic performance features become visible: curricular units credited, enrolled, evaluated, approved, grades, and units without evaluations ($21 + 6 = 27$ features).
- **Week 12 (Late-Semester):** The full feature set is unlocked, including second-semester academic performance (GPA, curricular unit metrics) and macroeconomic indicators (unemployment rate, inflation, GDP) ($27 + 9 = 36$ features).

Columns used to derive the target variable (`Target`, `G3`, `GradeClass`) are explicitly excluded from the feature set at all checkpoints to prevent target leakage.

D. Leakage-Free Preprocessing and Model Training

Algorithm 1 summarizes the per-checkpoint training protocol. The pipeline enforces four leakage-prevention mechanisms:

1) **Target isolation:** Source columns for the risk label are removed before feature selection.
2) **Post-split feature selection:** Mutual Information [28] scores are computed exclusively on the training partition; validation and test sets are subsetted to the same selected features, preventing selection bias leakage.

Algorithm 1 Per-Checkpoint Training Protocol

Require: Dataset D, checkpoint week $w \in \{4, 8, 12\}$
1: $\mathbf{X} \leftarrow$ Apply feature mask $\mathcal{F}_w$ to D
2: One-hot encode categorical features in $\mathbf{X}$
3: Split: $\mathbf{X}_{\text{train}}, \mathbf{X}_{\text{val}}, \mathbf{X}_{\text{test}}$ (70/15/15, stratified)
4: Compute MI scores $\text{MI}(\mathbf{X}_{\text{train}}, y_{\text{train}})$ on training data **only**
5: Select top-k features ($k = 15$) by MI ranking; subset all splits
6: Apply SMOTE ($k_{\text{neighbors}}$=5) to $(\mathbf{X}_{\text{train}}, y_{\text{train}})$ **only**
7: Fit StandardScaler on $\mathbf{X}_{\text{train}}$; transform all splits
8: Train models {LR, RF, XGBoost} on $\mathbf{X}_{\text{train}}$
9: Evaluate on $\mathbf{X}_{\text{test}}$ using F1-Macro, ROC-AUC, κ

3) **Post-split balancing:** SMOTE (original variant [12], $k_{\text{neighbors}}$=5) is applied exclusively to the training partition.
4) **Post-split scaling:** StandardScaler parameters are fit on the training partition only; validation and test sets are transformed using these learned parameters.

Feature selection retains the top-15 features per checkpoint based on MI ranking. Collinear features ($|r| > 0.85$) are removed prior to selection, eliminating 2–4 features per checkpoint depending on feature set size.

Models. Three classifiers are benchmarked:

- *Logistic Regression* (LR) [29]: baseline linear model (C=1.0, LBFGS solver, max 1000 iterations).
- *Random Forest* (RF) [5]: bagging ensemble (300 trees, max depth 15, balanced class weights).
- *XGBoost* [6]: gradient boosting (300 rounds, max depth 6, learning rate 0.1, subsample 0.8).

All hyperparameters are fixed *a priori* based on established literature recommendations; no hyperparameter tuning is performed on the validation set. The validation partition ($n = 15\%$) is reserved for monitoring generalization during development, while final metrics are reported exclusively on the held-out test partition ($n = 15\%$). To assess result stability, the full training and evaluation protocol is repeated across five random seeds (42, 123, 256, 512, 1024), with mean $\pm$ standard deviation reported.

E. Evaluation Metrics

We adopt F1-Macro as the primary metric because it equally weights precision and recall across both classes, penalizing poor minority-class performance [30]. Supplementary metrics include accuracy, macro-averaged precision and recall, ROC-AUC, and Cohen's κ [31].

F. Three-Tier Risk Stratification and Explainability

Predicted risk probabilities are mapped to a three-tier intervention framework:

$$\text{Tier}(\mathbf{x}) = \begin{cases} \text{Low Risk} & \text{if } P(\text{risk}|\mathbf{x}) < 0.30 \\ \text{Medium Risk} & \text{if } 0.30 \leq P(\text{risk}|\mathbf{x}) \leq 0.60 \\ \text{High Risk} & \text{if } P(\text{risk}|\mathbf{x}) > 0.60 \end{cases} \tag{1}$$

For all students classified as Medium or High risk, SHAP TreeExplainer [22] generates local feature attributions. Global feature importance is computed as the mean absolute SHAP value across all test samples: $\bar{\phi}_j = \frac{1}{N}\sum_{i=1}^{N} |\phi_j^{(i)}|$. Individual waterfall plots decompose each prediction into per-feature contributions, providing counselors with the specific factors driving a student's risk classification.

IV. Experimental Results

All experiments are conducted on the UCI Dropout dataset (n = 4,424), with the train/val/test split fixed at 70/15/15 using stratified sampling (random seed = 42). The test set ($n = 664$) comprises 451 non-risk and 213 at-risk students, yielding a majority-class baseline accuracy of 67.9%. Results are reported on the held-out test set; variance estimates are obtained via a five-seed stability analysis (Section IV-D).

A. Model Performance at Checkpoint 12

Table I presents the full evaluation metrics for all three models at the Week 12 checkpoint, where the complete feature set is available.

TABLE I: Model Performance at Checkpoint 12 (Test Set)

Model	Acc.	Prec.	Rec.	F1-M	AUC	κ
Logistic Reg.	0.858	0.836	0.845	0.840	0.922	0.680
Random Forest	0.869	0.848	0.854	0.851	**0.925**	0.702
XGBoost	**0.870**	**0.851**	**0.853**	**0.852**	0.920	**0.704**

XGBoost achieves the highest F1-Macro (0.852) and Cohen's κ (0.704), indicating substantial agreement beyond chance [31]. All models substantially outperform the 67.9% majority-class baseline, with even the linear baseline (LR) exceeding 0.84 F1-Macro, demonstrating that the fully leakage-free pipeline—including post-split feature selection—produces consistently strong results. Notably, Random Forest achieves the highest ROC-AUC (0.925), slightly exceeding XGBoost (0.920); this inversion occurs because ROC-AUC evaluates the full ranking of predicted probabilities whereas F1-Macro evaluates performance at a fixed decision threshold, and XGBoost's sharper probability calibration favors the latter.

Fig. 2 displays the ROC curves for the three models at Checkpoint 12, confirming that ensemble methods and the linear baseline achieve comparable discriminative ability on this feature set.

Fig. 3 presents the confusion matrices, illustrating the error distribution across risk categories for each model.

B. Temporal Checkpoint Analysis

Table II presents the F1-Macro progression across all three temporal checkpoints, and Fig. 4 visualizes this evolution.

Three observations emerge from the temporal analysis:

1) **Week 4 baseline:** Using only enrollment and demographic features, all models achieve F1-Macro in the range 0.702–0.735, substantially exceeding the 67.9% majority-class baseline. This represents the lower bound of predictive capability using static pre-admission data alone.

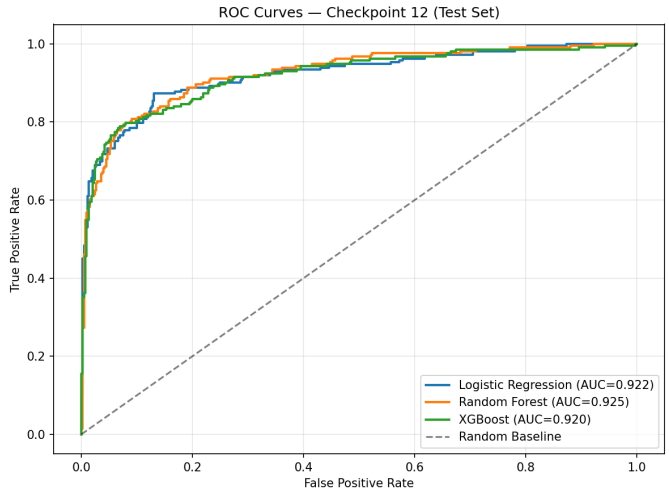


Fig. 2: ROC curves at Checkpoint 12 (test set). All models achieve AUC $\geq$ 0.920, indicating strong discriminative performance.

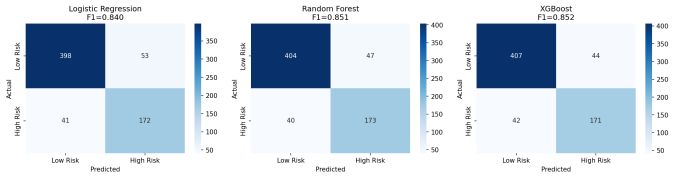


Fig. 3: Confusion matrices at Checkpoint 12 (test set). XGBoost achieves the highest F1-Macro (0.852) with balanced precision and recall across both classes.

TABLE II: F1-Macro Across Temporal Checkpoints (Test Set)

Model	Week 4	Week 8	Week 12
Logistic Regression	0.735	0.806	0.840
Random Forest	0.715	**0.819**	0.851
XGBoost	0.702	0.813	**0.852**

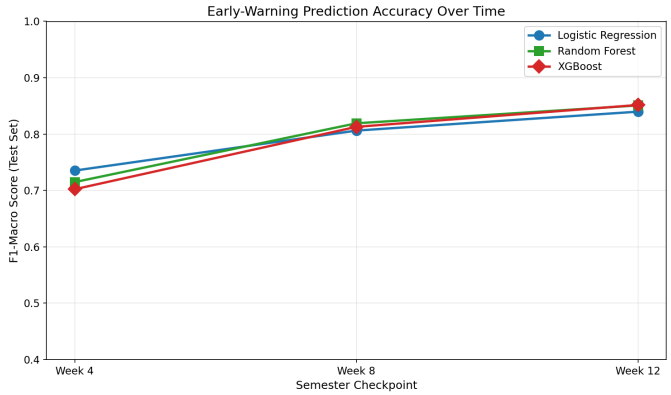


Fig. 4: F1-Macro evolution across temporal checkpoints. The transition from Week 4 to Week 8 produces the largest performance gain ($\Delta \approx +0.08$–0.11), coinciding with the introduction of first-semester academic features.

2) **Week 4 $\rightarrow$ Week 8 transition:** The introduction of first-semester academic performance features at Week 8 produces the largest single-step improvement, with F1-Macro increasing by 0.08–0.11 points across all models. This confirms that behavioral academic signals carry substantially more predictive power than static demographic variables.
3) **Week 8 as the operational optimum:** While Week 12 yields the highest absolute performance, Week 8 models achieve F1-Macro $\geq$ 0.806 while providing eight additional weeks of lead time before semester end. This represents the most favorable point on the accuracy–intervention tradeoff curve for practical deployment.

C. SHAP-Based Explainability

1) Global Feature Importance: Fig. 5 presents the global SHAP feature importance for the XGBoost model at Checkpoint 12, computed as the mean absolute SHAP value across all test samples. Fig. 6 extends this analysis with a directional beeswarm plot showing how each feature's value affects risk prediction.

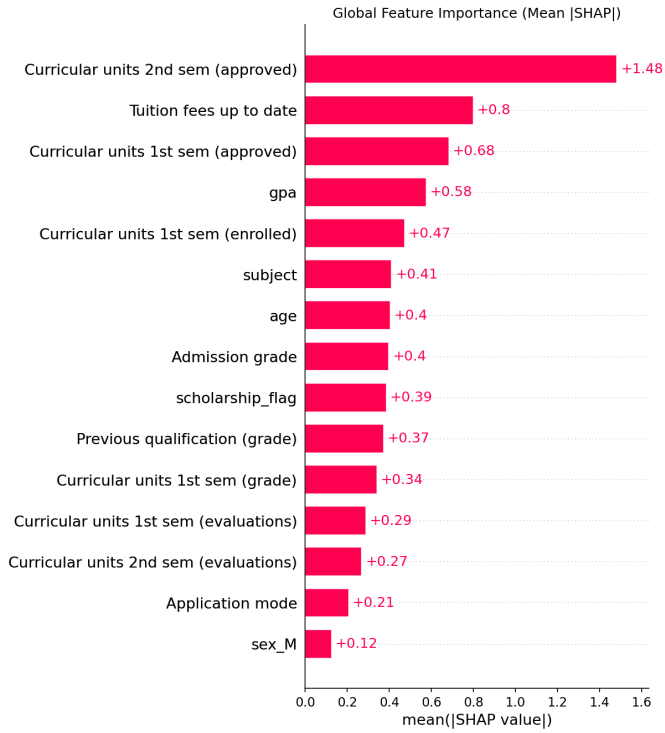


Fig. 5: Global SHAP feature importance (mean $|\phi_j|$). Second-semester curricular performance and GPA dominate, followed by first-semester indicators and enrollment features.

The global analysis reveals that academic performance metrics—particularly second-semester curricular unit outcomes and GPA—are the dominant predictors, followed by first-semester indicators. Enrollment-time features (age, application mode) contribute measurably but with lower magnitude, consistent with the temporal checkpoint results in Section IV.

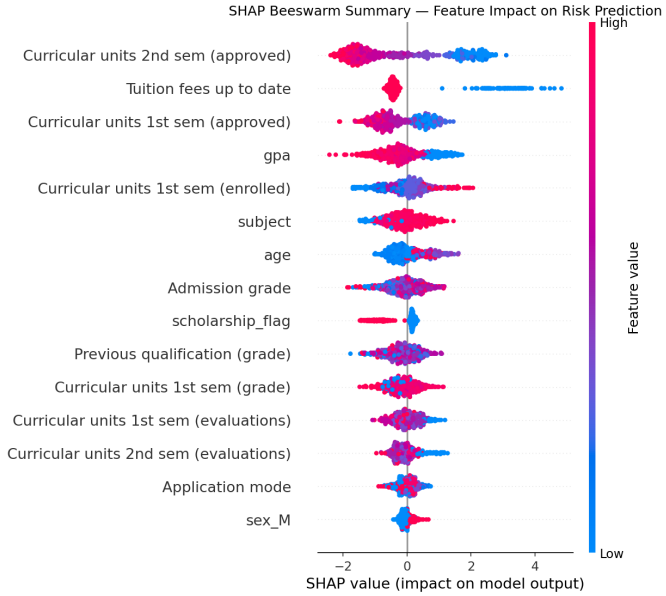
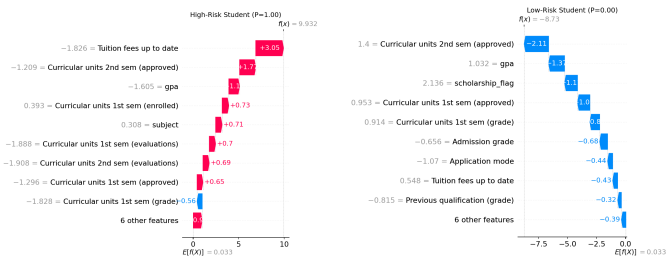


Fig. 6: SHAP beeswarm summary. Red (high feature value) and blue (low value) dots reveal the direction of each feature's impact on risk prediction. Low GPA and high curricular unit failures consistently increase predicted risk.

2) Individual Student Explanations: Fig. 7 presents localized SHAP waterfall plots for a high-risk and a low-risk student from the test set.



(a) High-risk student: low approved curricular units and poor GPA drive the elevated risk score.

(b) Low-risk student: strong GPA and high curricular unit completion serve as protective factors.

Fig. 7: Localized SHAP waterfall plots. Each bar quantifies a single feature's contribution to the individual prediction, enabling counselors to identify specific intervention targets per student.

These individual explanations transform abstract risk scores into actionable guidance. For example, a counselor reviewing the high-risk profile in Fig. 7(a) can identify that the student's primary risk drivers are poor curricular performance and low GPA, enabling targeted academic support rather than generic advising.

D. Stability Analysis and Computational Cost

To assess the robustness of reported metrics, the full training and evaluation protocol is repeated across five random seeds (42, 123, 256, 512, 1024). Table III presents the mean $\pm$ standard deviation of F1-Macro at each checkpoint.

TABLE III: Multi-Seed Stability Analysis: F1-Macro (Mean $\pm$ Std, n=5 Seeds)

Model	Week 4	Week 8	Week 12
Logistic Reg.	0.735 ± 0.000	0.806 ± 0.000	0.840 ± 0.000
Random Forest	0.721 ± 0.008	0.821 ± 0.002	0.852 ± 0.002
XGBoost	0.698 ± 0.001	0.820 ± 0.004	0.847 ± 0.005

All models exhibit low variance across seeds ($\sigma \leq 0.008$), confirming the stability of reported results. Logistic Regression, being deterministic given fixed data splits, shows zero variance. The 5-fold cross-validation F1-Macro on training data further corroborates test-set performance: XGBoost achieves 0.886 ± 0.008, Random Forest 0.882 ± 0.008, and Logistic Regression 0.860 ± 0.008 at Week 12.

Table IV summarizes computational costs. All experiments are conducted on a consumer-grade workstation (Intel Core i7, 16 GB RAM, no GPU acceleration).

TABLE IV: Training Time per Model at Checkpoint 12

Model	Train (s)	Inference (s)
Logistic Regression	0.02	< 0.01
Random Forest	0.87	< 0.01
XGBoost	0.54	< 0.01

The sub-second training times confirm that ProGrade is deployable on institutional hardware without specialized infrastructure. End-to-end pipeline execution (feature engineering through evaluation across all three checkpoints) completes in under 30 seconds.

V. Discussion

A. Addressing the Research Questions

RQ1 (Architecture and Reproducibility): The Medallion pipeline provides three key advantages over flat-file approaches. First, immutable Bronze storage preserves raw data provenance for audit. Second, the Silver star-schema enforces referential integrity and enables consistent cross-dataset joins. Third, the Gold data mart decouples analytical outputs from the ML pipeline, allowing BI dashboards and counselor reports to operate independently of model retraining cycles. These architectural decisions directly support reproducibility, as any downstream analysis can be traced to its source records.

RQ2 (Optimal Checkpoint): The temporal analysis reveals Week 8 as the most operationally valuable checkpoint. The Week 4 $\rightarrow$ Week 8 transition yields the largest marginal gain (ΔF1 $\approx$ +0.08–0.11), driven by the availability of first-semester academic features. The subsequent Week 8 $\rightarrow$ Week 12 gain is more modest (ΔF1 $\approx +0.03$), while the lead time decreases by four weeks. For institutions prioritizing early intervention, Week 8 provides an F1-Macro exceeding 0.80 with eight weeks remaining for counselor-led remediation. The five-seed stability analysis (Table III) confirms that this

conclusion is robust across random seeds ($\sigma \leq 0.004$ at Week 8).

RQ3 (Explainability): The combination of global SHAP importance rankings (Fig. 5) and individual waterfall plots (Fig. 7) addresses the trust deficit identified in prior work [15]. Rather than presenting a binary risk label, ProGrade provides counselors with a ranked list of contributing factors per student, mapped to the three-tier classification in Eq. 1. This supports differentiated interventions—e.g., academic tutoring for students with low GPA attribution versus attendance monitoring for absence-driven risk.

B. Comparison with Prior Work

Table V positions ProGrade relative to recent studies on the same or comparable datasets.

TABLE V: Comparison with Related Work on Student Risk Prediction

Study	Metric	Score	Temporal	XAI
Realinho et al. [1]	Accuracy	0.79	No	No
Martins et al. [21]	F1-Macro	0.83	Yes	No
Ahmad et al. [32]	Accuracy	0.88	No	No
Alwadi et al. [16]	Accuracy	0.85	No	Yes
ProGrade (ours)	**F1-Macro**	$\mathbf{0.852 \pm 0.005}$	**Yes**	**Yes**

Note: Direct numerical comparison across studies is approximate, as reported metrics differ (Accuracy vs. F1-Macro). ProGrade reports F1-Macro with five-seed variance estimates to ensure methodological transparency.

ProGrade is, to our knowledge, one of the first studies to unify a Medallion warehouse architecture, systematic temporal checkpoint evaluation, fully leakage-free preprocessing (including post-split feature selection), and SHAP-based per-student explanations within a single framework.

C. Limitations and Future Work

Several limitations warrant discussion:

1) **Simulated temporal masking:** The checkpoint protocol simulates temporal availability by masking features in a static dataset. Authentic deployment requires integration with institutional Learning Management Systems (LMS) for real-time data feeds.
2) **Single-institution datasets:** While three datasets are unified in the warehouse, they originate from similar educational contexts (Portuguese institutions). Cross-institutional transfer learning [24] remains unexplored.
3) **Binary risk classification:** The current pipeline predicts a binary outcome (at-risk / not-at-risk). Multi-class prediction (Dropout / Enrolled / Graduate) [21] may provide more granular intervention guidance.
4) **Feature modality:** The current system operates exclusively on structured tabular data. Future work could integrate NLP-based analysis of student communication logs and LMS engagement traces as supplementary behavioral signals [23].
5) **Fairness analysis:** We do not formally assess model fairness across demographic subgroups. Given the inclusion of sensitive attributes (gender, socioeconomic status), future work should employ established fairness metrics—including equalized odds, demographic parity, and calibration across groups—as a prerequisite for institutional deployment [24].

VI. Conclusion

This paper presented ProGrade, an end-to-end early warning framework that addresses four persistent gaps in educational data mining: the absence of reproducible data architectures, data leakage in preprocessing, temporal mismatch between prediction and intervention, and insufficient model transparency. By integrating a Medallion data warehouse (Bronze $\rightarrow$ Silver Star-Schema $\rightarrow$ Gold Data Mart) with fully leakage-free preprocessing (including post-split feature selection), temporal checkpoint modeling, and SHAP-based explainability, ProGrade provides a principled foundation for institutional deployment.

Our experimental evaluation on the UCI Dropout dataset ($n = 4{,}424$) demonstrates that XGBoost achieves an F1-Macro of 0.852 ± 0.005 and ROC-AUC of 0.920 at Week 12 (five-seed stability analysis), with Week 8 models attaining F1-Macro ≥ 0.80 while preserving eight weeks of intervention lead time. All models exhibit low variance across seeds ($\sigma \leq 0.008$), confirming result robustness, and sub-second training times confirm deployability on consumer-grade hardware. SHAP TreeExplainer generates transparent, per-student explanations that enable academic counselors to identify specific risk drivers and design targeted intervention strategies.

Future work will focus on real-time LMS integration, multi-class outcome prediction, cross-institutional transfer learning, and formal fairness auditing using equalized odds and demographic parity metrics, to advance ProGrade toward production deployment in diverse educational settings.

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